AI Water Quality Forecasting System

CSE276: Artificial Intelligence Foundations | CA1 Project-Based Assignment | Project P18 (AI Water Quality Forecasting System) | Team ZJ18: Prince Sahu (12508888) and Ridham Dadwal (12514189) | Lovely Professional University

1. Introduction and Problem Statement

Water quality changes with season, rainfall, temperature and pollution load. Authorities who monitor rivers need to know not only what the water quality is today but also what it is likely to be soon, so that they can act in time. This project addresses problem statement P18, mapped to SIH 2024 SIH1694 (real-time river water quality forecasting using AI-enabled decision support), and asks for a multivariate forecasting and decision support system (DSS) built on public water-quality measurements.

We built a working system with two parts:

- **Part A (real data):** next-year Water Quality Index (WQI) forecasting for Indian river monitoring stations, using a public dataset of yearly station averages (2003-2014).
- **Part B (synthetic data):** a short-term (7-day ahead) forecasting and alert prototype on a documented synthetic daily dataset, because we did not find a public daily river dataset to use.

Both parts feed a Streamlit demo application with decision-support alerts.

2. Objectives

1. Clean and prepare multivariate water-quality data (pH, dissolved oxygen, BOD, conductivity, temperature, nitrate, coliform, turbidity).
2. Compute a Water Quality Index (WQI) that summarises the parameters in one number and one category.
3. Forecast the WQI with statistical and machine-learning models and compare them against a naive baseline using a time-based test.
4. Convert forecasts into decision-support alerts and present them in an interactive demo.
5. Document dataset limitations honestly, including what the synthetic data can and cannot show.

3. Datasets

3.1 Real data (Part A)

Item	Detail
Source	Kaggle dataset "Indian water quality data" (file water_dataX.csv), described by its author as combined and cleaned historical water-quality data from Indian government websites. Link and licence: Dataset page: https://www.kaggle.com/datasets/anbarivan/indian-water-quality-data (author Anbarivan.N.L, accessed 26 October 2026). We could not read a licence statement from the page we retrieved, so the data were used only for academic, non-commercial coursework; the original source is Indian government websites.
Content	Yearly average readings per monitoring station and year, 2003-2014
Size	1,988 usable rows after removing rows with no station identity; 439 stations
Parameters	Temperature, dissolved oxygen (DO), pH, conductivity, biochemical oxygen demand (BOD), nitrate+nitrite, fecal coliform, total coliform
Not available	Turbidity (listed in the problem statement) is not in this dataset
Geography	20 states and union territories; 11 rows have no state information (Unknown)

Data-quality issues found and handled:

- Many values are missing and stored as the text "NAN". These were converted to missing values and were never replaced by zero.
- In about 6% of rows the station code is "NAN". Treating them as one station would have merged unrelated places, so the location name was used as the station identifier, and rows with neither were dropped.

- The STATE column sometimes contained a location text (for example, a river location followed by a state name) and had several spellings of the same state. The state was recovered from the end of the text where possible, spellings were unified (for example TAMILNADU to TAMIL NADU, ORISSA to ODISHA), and a state known for a station in one year was applied to its other years. After this, 203 distinct raw state values reduced to 20 states and union territories.
- Many stations have little history: on average 4.5 years of data per station, a median of 3, and at least a quarter of stations have only one year.

3.2 Synthetic data (Part B)

The problem statement needs forecasting, and a daily time series is the most natural setting for it. We could not locate a public daily river dataset suitable for this assignment, so we generated a documented synthetic dataset to build and demonstrate a short-term forecasting pipeline.

Item	Detail
Period	1 January 2019 to 31 December 2024 (2,192 daily records), one station (Station_A)
Parameters	Temperature, pH, dissolved oxygen, turbidity, conductivity, BOD
Design	Annual temperature cycle, monsoon-driven rise in turbidity and BOD and fall in conductivity and pH (peak around July to September), temporally correlated noise, DO inversely related to temperature and turbidity
Imperfections added on purpose	About 3% missing values per column, and 15 impossible readings (pH 14.9, turbidity 999, DO -5) to exercise the cleaning step

Because the series was generated from smooth patterns, results on it are expected to look better than results on real data. It is used as a prototype and for validation of the pipeline, not as evidence of real-world accuracy.

4. Methodology

The system has two pipelines that share one application: Part A on real yearly data and Part B on synthetic daily data (Figure 1).

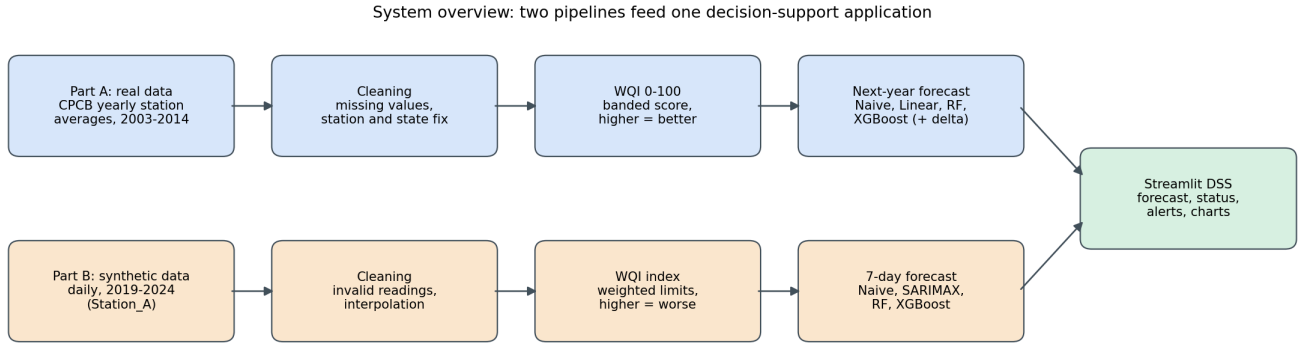


Figure 1. System overview: two pipelines, one for real yearly data and one for synthetic daily data, feed one decision-support application.

4.1 Cleaning

Values outside physically possible ranges were set to missing (for example pH outside 4-10 in the synthetic data, DO below 0). In the synthetic data, series were resampled to daily frequency and gaps were filled by time interpolation. In the real data, missing values were kept as missing and handled by the WQI calculation and by imputation inside the models (see below).

4.2 WQI for the real data (0-100, higher = better)

Each parameter is converted to a sub-score (100, 80, 60, 40 or 0) using standard-style bands, then a weighted average is taken. Missing parameters are left out and the weights of the available ones are renormalised; a station-year needs at least four of the six parameters to receive a WQI. The bands follow the CPCB-style approach used in a public Kaggle notebook on this same dataset (Anbarivan N.L), with some thresholds adjusted by us (pH, BOD and conductivity). The top band is consistent with the CPCB designated-best-use criteria for a drinking-water source (class A: pH 6.5 to 8.5, DO at least 6 mg/l, BOD at most 2 mg/l, total coliform at most 50 MPN/100 ml), except that our BOD limit is looser (3 mg/l) and our coliform limit is tighter (5 MPN/100 ml).

Parameter	Weight	Bands for scores 100 / 80 / 60 / 40
pH	0.165	6.5-8.5 / 6.0-6.5 or 8.5-9.0 / 5.5-6.0 or 9.0-9.5 / 5.0-5.5 or 9.5-10
Dissolved oxygen (mg/l)	0.281	at least 6 / at least 5 / at least 4 / at least 3
BOD (mg/l)	0.234	up to 3 / 6 / 10 / 30
Total coliform (MPN/100ml)	0.281	up to 5 / 50 / 500 / 10,000
Conductivity (umhos/cm)	0.009	up to 750 / 1,500 / 2,250 / 3,000
Nitrate (mg/l)	0.028	up to 20 / 50 / 100 / 200

Categories: Excellent (90 or more), Good (70-90), Medium (50-70), Bad (25-50), Very Bad (below 25). Result on 1,955 station-years: Good 1,434, Excellent 219, Medium 195, Bad 93, Very Bad 14.

4.3 WQI for the synthetic data (higher = worse)

A weighted arithmetic index was used: for each parameter, Qi = |Vi - Videal| / (Si - Videal) x 100 with weight Wi = 1/Si, and WQI = sum(Wi x Qi) / sum(Wi). Standards: pH 8.5 (ideal 7.0), DO 5.0 (ideal 14.6), turbidity 100, conductivity 750, BOD 3.0. Categories: up to 50 Excellent, 50-100 Good, 100-200 Poor, 200-300 Very Poor, above 300 Unsafe. Over the synthetic period the WQI had a mean of about 94.7, a minimum of 60.2 and a maximum of 159.9 (1,508 Good days, 684 Poor days).

Important: the two WQI scales point in opposite directions (higher is better in Part A, higher is worse in Part B). Every figure, table and application screen in this project states the scale it uses.

4.4 Forecasting setup, Part A (next-year WQI, real data)

- **Task:** use a station's readings and WQI in year t to predict its WQI in year t+1. A pair exists only when the station has data in both years.
- **Features:** temperature, DO, pH, conductivity, BOD, nitrate, log of total coliform, log of fecal coliform, current WQI, previous-year WQI (when available) and the change between them.
- **Split (time-based, no random split):** training on pairs with base year 2003-2011 (898 pairs); testing on base years 2012-2013, i.e. forecasting 2013 and 2014 (478 pairs).

- **Models:** (1) Naive baseline: next year equals this year; (2) Linear regression; (3) Random forest (400 trees, minimum leaf size 3); (4) XGBoost (300 trees, depth 3); plus a **delta** version of the linear, random forest and XGBoost models that predict the change in WQI and add it to the current WQI (the linear delta model uses ridge regularisation). Missing feature values are imputed with the training median for the linear and random forest models, while XGBoost handles missing values natively.
- **Metrics:** MAE, RMSE, R-squared and category accuracy (share of forecasts that land in the correct WQI category).

4.5 Forecasting setup, Part B (7-day ahead, synthetic data)

- **Features:** lags (0, 1, 2, 3, 7, 14 days) and a 7-day rolling mean of every parameter and of the WQI, plus sine and cosine of the day of year.
- **Split:** first 80% of the series for training, last 20% for testing, with a 7-row gap to avoid leakage; one forecast every 7 days in the test period (63 forecasts).
- **Models:** Naive (today's WQI), SARIMAX with order (2,0,1) and annual seasonal terms, random forest (300 trees) and XGBoost (400 trees, depth 4).
- **Metrics:** MAE, RMSE, MAPE and category accuracy.

4.6 Decision-support application

A Streamlit application with two tabs. The real-data tab lets a user choose a state, station, base year and model, and shows the current WQI, the forecast for the next year, the actual value when it exists in the data, a coloured status message (acceptable, treatment recommended, or alert), the station's WQI history with forecast, the readings, the model comparison table and the all-station yearly trend. The synthetic tab shows the 7-day forecast with the same kind of status and charts.

Figures 2 to 4 show the application: the lower half of the real-data tab and the synthetic-data tab.

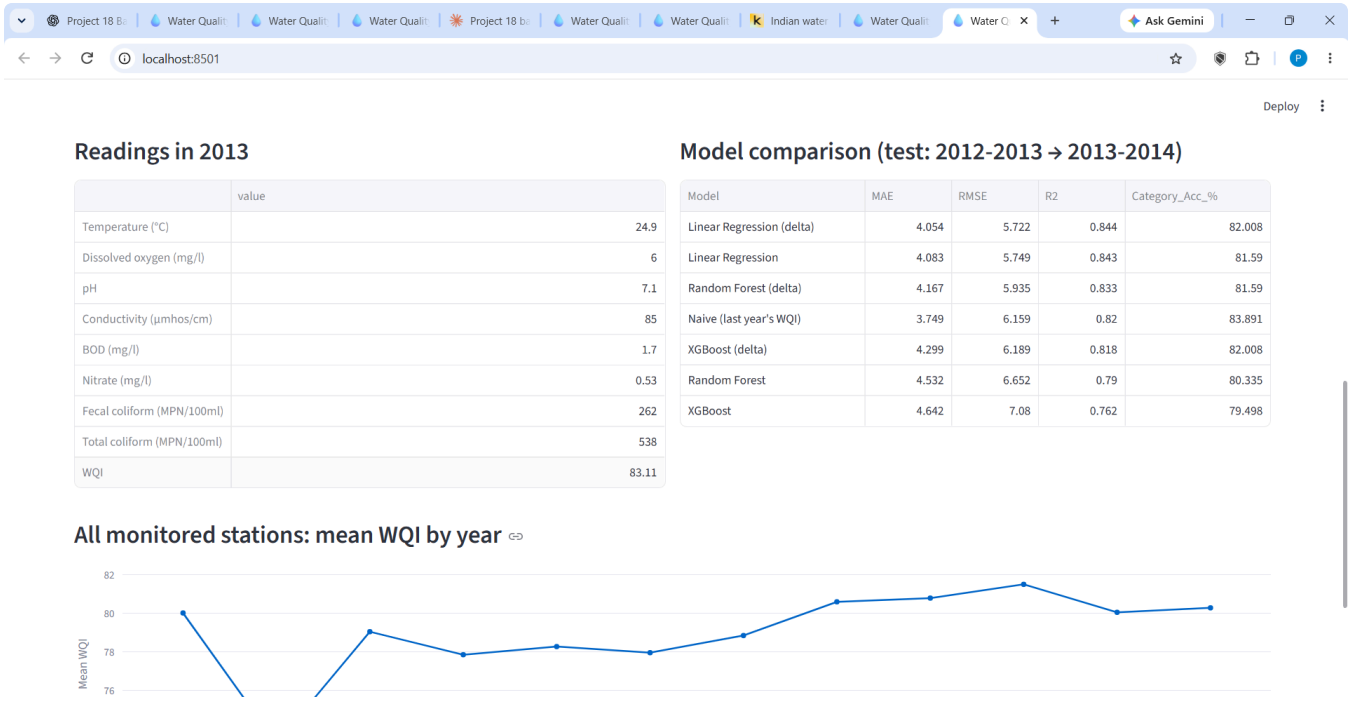


Figure 2. Real-data tab (lower half): readings of the selected station, model comparison table and mean WQI of all stations by year (real WQI: higher = better).

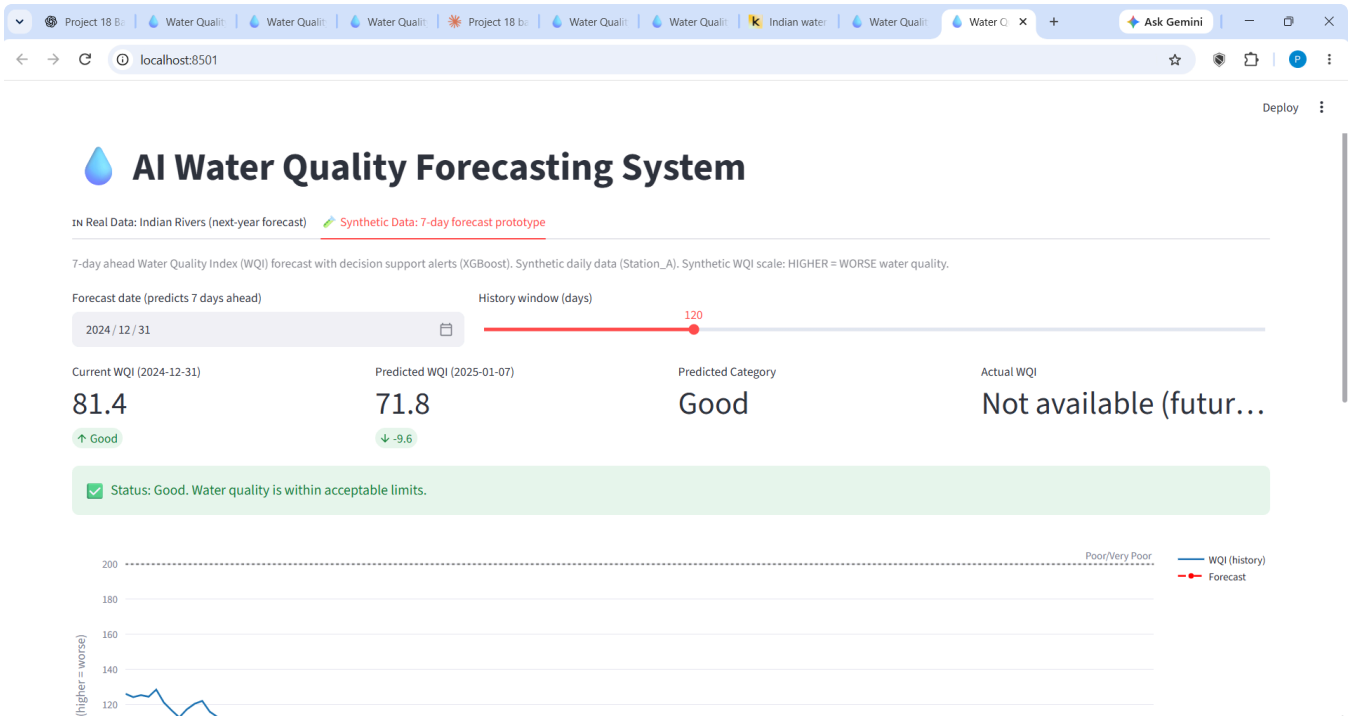
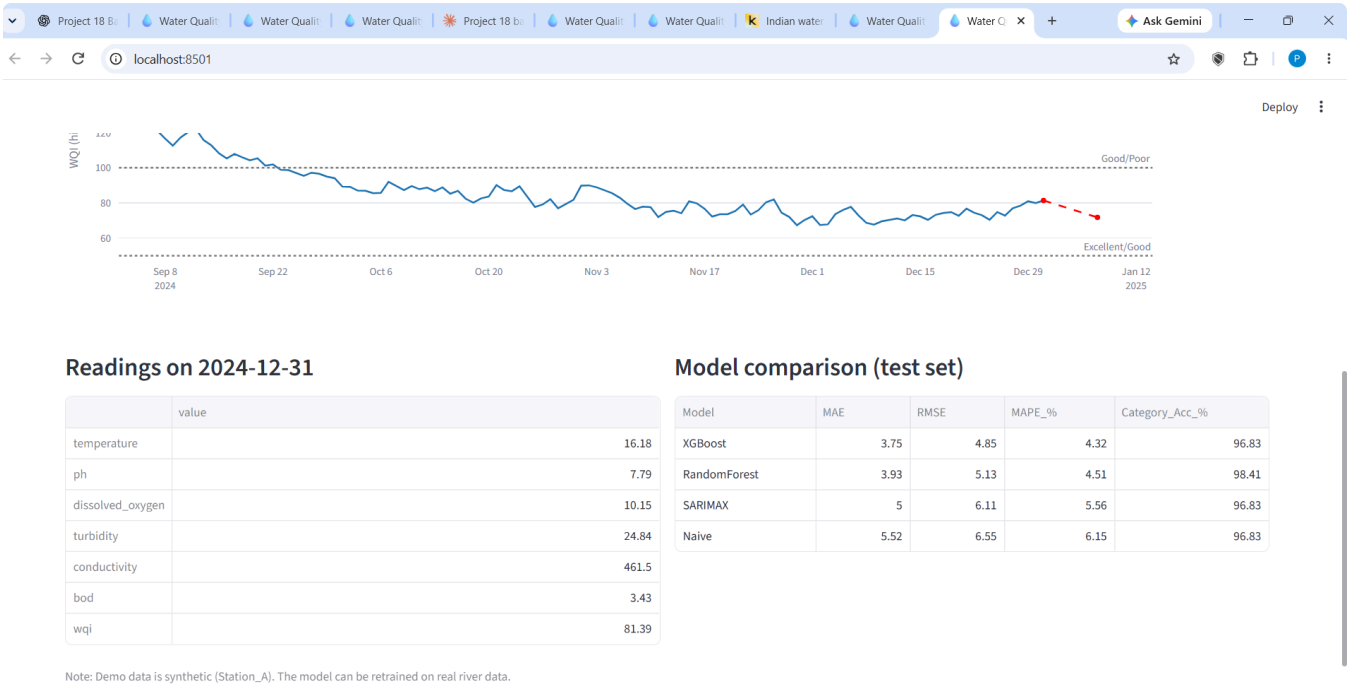


Figure 3. Synthetic-data tab: current and predicted WQI, predicted category and status message (synthetic WQI: higher = worse).



Readings on 2024-12-31

	value
temperature	16.18
ph	7.79
dissolved_oxygen	10.15
turbidity	24.84
conductivity	461.5
bod	3.43
wqi	81.39

Model comparison (test set)

Model	MAE	RMSE	MAPE_%	Category_Acc_%
XGBoost	3.75	4.85	4.32	96.83
RandomForest	3.93	5.13	4.51	98.41
SARIMAX	5	6.11	5.56	96.83
Naive	5.52	6.55	6.15	96.83

Note: Demo data is synthetic (Station_A). The model can be retrained on real river data.

Figure 4. Synthetic-data tab (lower part): WQI history with the 7-day forecast, readings and model comparison.

5. Results

5.1 Part A: real Indian river data (next-year WQI, test years 2013-2014)

Model	MAE	RMSE	R-squared	Category accuracy (%)
Linear regression (delta)	4.054	5.722	0.844	82.01
Linear regression	4.083	5.749	0.843	81.59
Random forest (delta)	4.167	5.935	0.833	81.59
Naive (last year's WQI)	3.749	6.159	0.820	83.89
XGBoost (delta)	4.299	6.189	0.818	82.01
Random forest	4.532	6.652	0.790	80.34
XGBoost	4.642	7.080	0.762	79.50

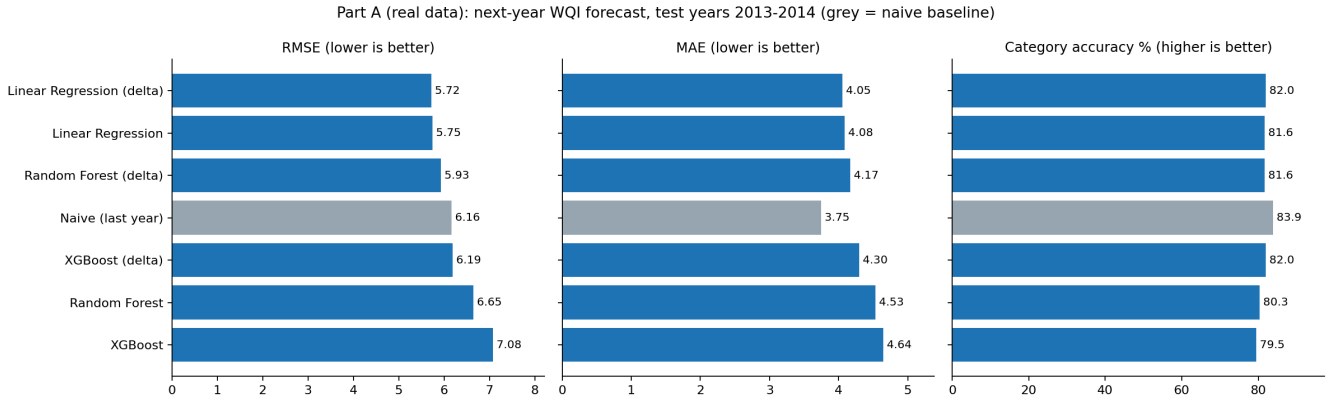


Figure 5. Part A model comparison (bars start at zero; the grey bar is the naive baseline).

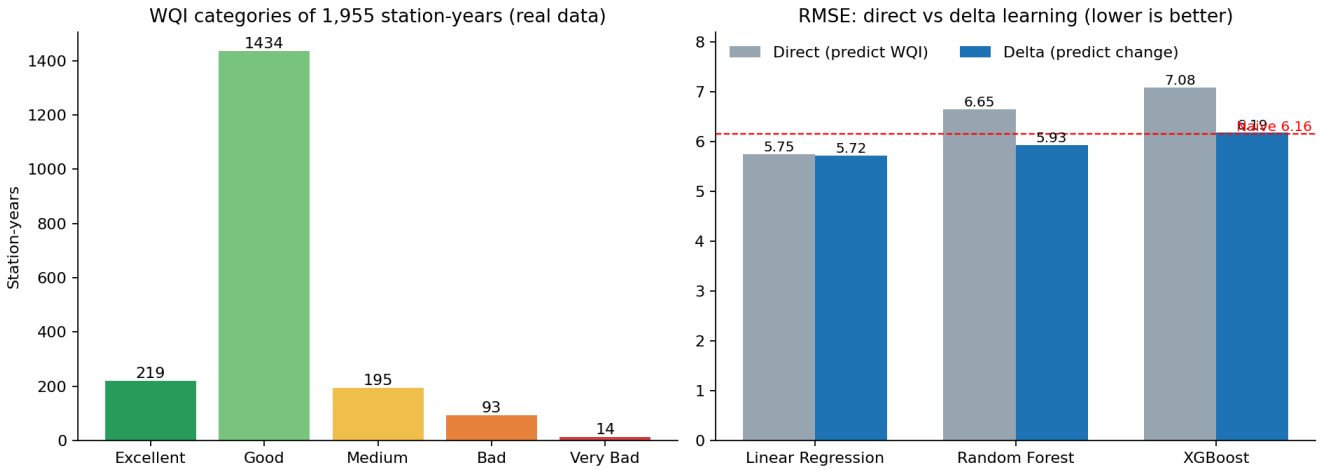


Figure 6. Left: WQI categories of the 1,955 station-years. Right: RMSE of direct and delta models against the naive baseline (red line).

5.2 Part B: synthetic daily data (7-day ahead, 63 forecasts)

Model	MAE	RMSE	MAPE (%)	Category accuracy (%)
XGBoost	3.75	4.85	4.32	96.83
Random forest	3.93	5.13	4.51	98.41
SARIMAX	5.00	6.11	5.56	96.83
Naive	5.52	6.55	6.15	96.83

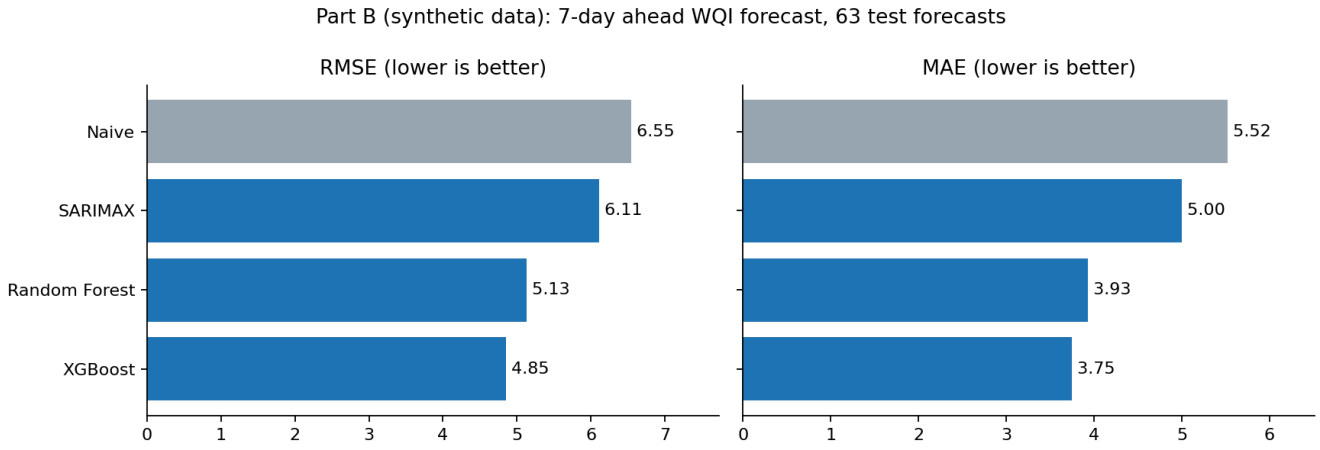


Figure 7. Part B model comparison (bars start at zero; the grey bar is the naive baseline).

The exploratory analysis of the synthetic series shows the intended monsoon pattern (Figures 8 to 11): turbidity, BOD and the WQI rise from July to September, conductivity and pH fall, and dissolved oxygen moves opposite to temperature.

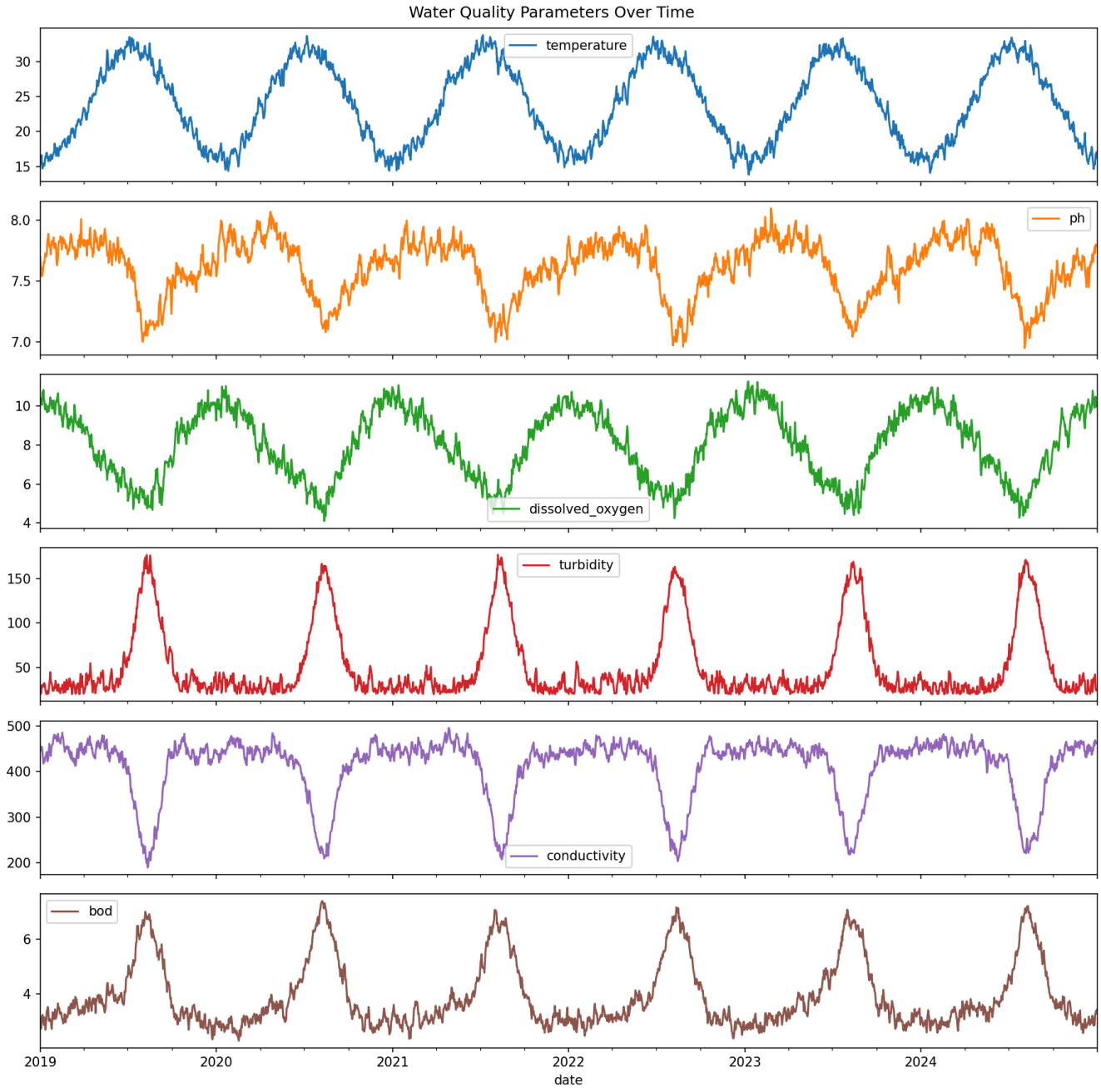


Figure 8. Synthetic parameters over time, 2019 to 2024.

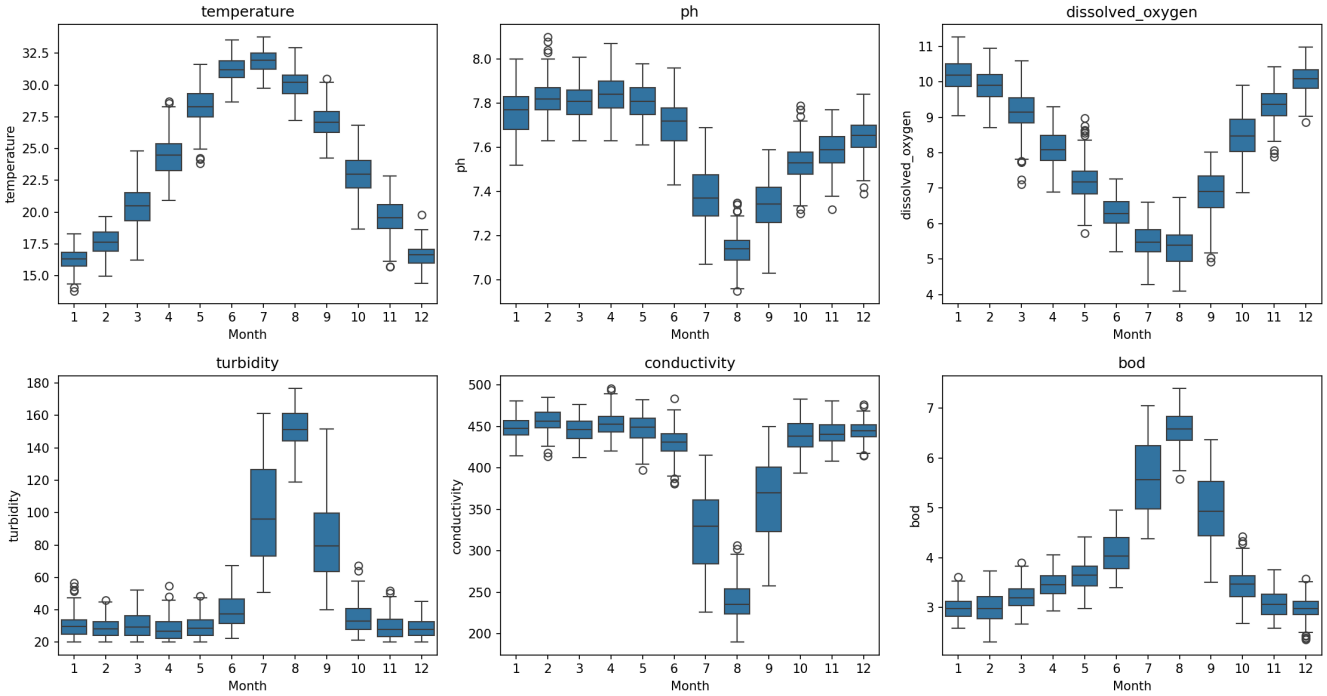


Figure 9. Monthly distribution of each parameter.

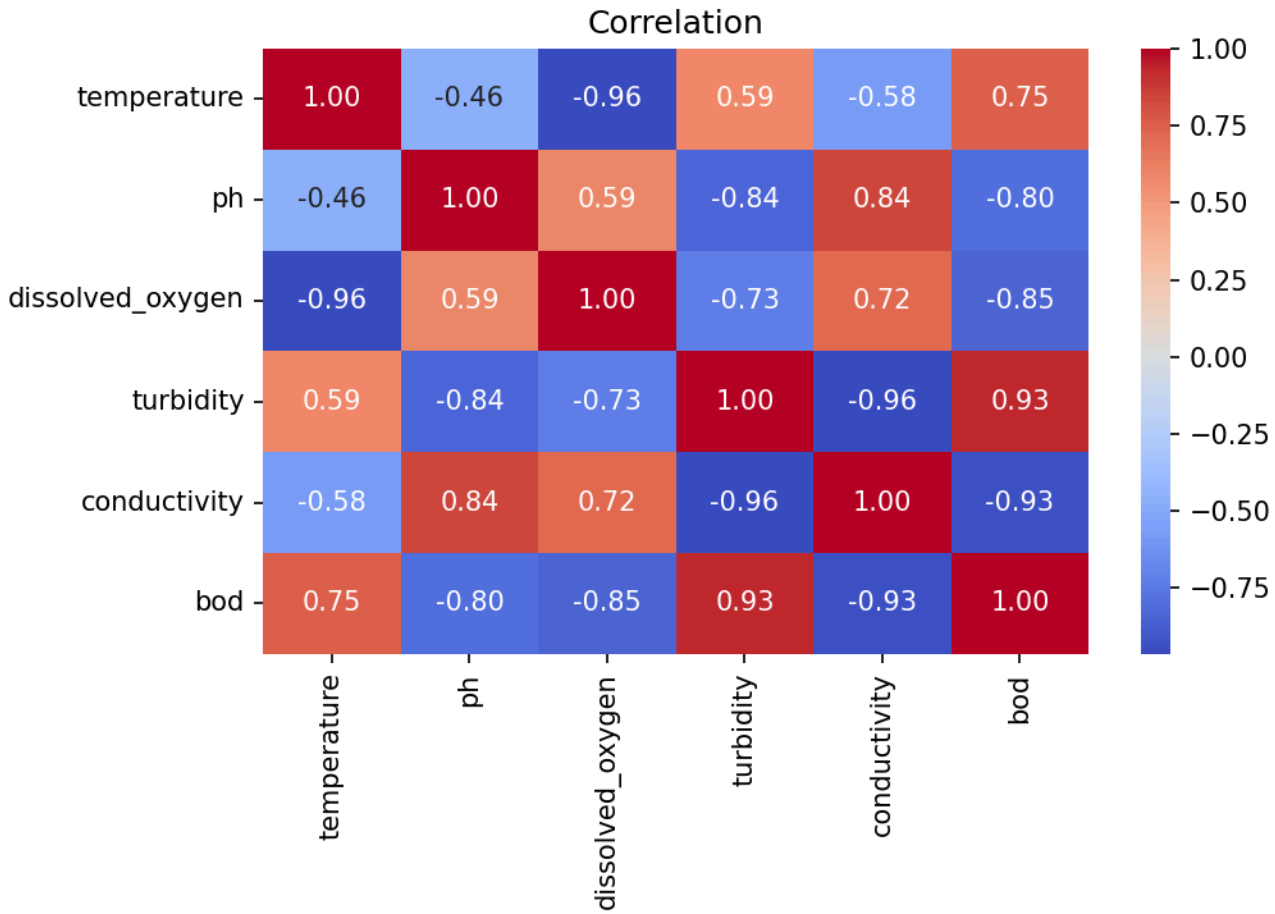


Figure 10. Correlation between the synthetic parameters.

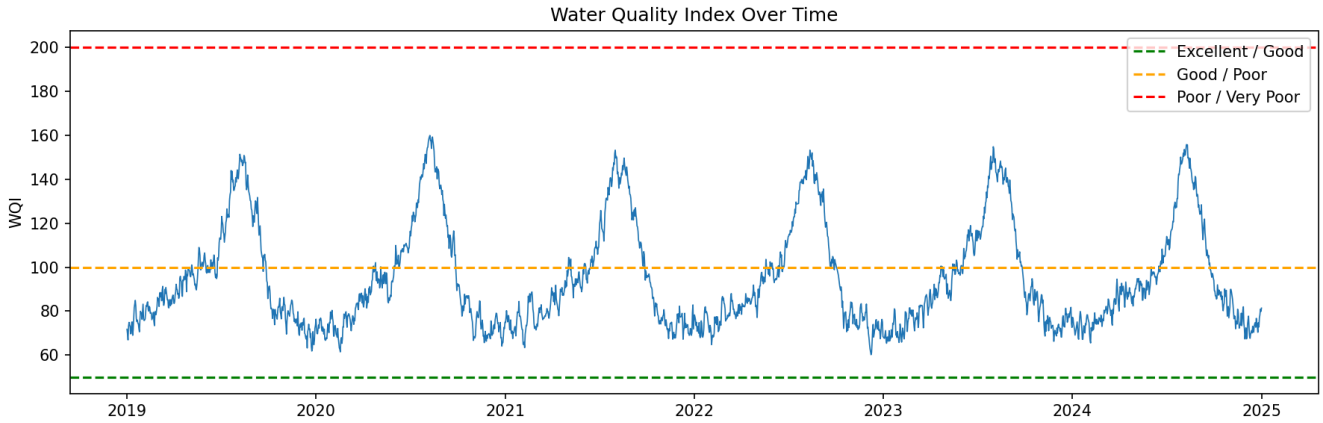


Figure 11. Synthetic WQI over time with the category thresholds (higher = worse).

6. Discussion

Real data. The linear delta model gave the lowest RMSE (5.72) and the highest R-squared (0.844), about 7% lower RMSE than the naive baseline (6.16). The random forest delta model also beat the baseline on RMSE (5.94). However, the naive baseline still had the lowest MAE (3.75) and the highest category accuracy (83.9%). In plain terms, the models reduce large errors somewhat, but for a typical station "next year will look like this year" is already hard to beat. Predicting the change in WQI instead of its level helped the tree models clearly: RMSE of the random forest fell from 6.65 to 5.94 and that of XGBoost from 7.08 to 6.19, because the model starts from the naive forecast and only has to learn a correction.

The improvements over the baseline are small and are measured on 478 test pairs, and we did not run a statistical significance test, so we describe them as modest gains rather than a decisive win. Likely reasons for the limited gain are the yearly averaging (which hides seasonal and event-driven variation), the short history of most stations, the stepwise nature of the banded WQI and the many missing values.

Synthetic data. The tree models beat both the naive baseline and SARIMAX in RMSE (XGBoost 4.85 against 6.55 for the naive baseline). These numbers are better than those of the real data mainly because the synthetic series is smooth and strongly seasonal by construction. They show that the pipeline works end to end and that lag features capture seasonal dynamics; they do not show how well the models would do on real daily river data.

7. Limitations and Future Work

- The real dataset contains yearly averages, not daily or monthly readings, so a true short-term forecast cannot be validated on real data with it.
- Turbidity, which the problem statement lists, is absent from the real dataset.
- Many values are missing; at least a quarter of stations have only one year of data; 11 rows have no state.
- The WQI uses banded sub-scores whose thresholds were adapted by us from public practice; results depend on those choices.
- No significance testing or cross-station cross-validation was done; the test set covers two years.
- The synthetic dataset is idealised and must not be read as real performance.

Future work: obtain daily or monthly station data (for example from CPCB or WRIS portals, subject to access and licence), retrain the Part B pipeline on it, add spatial relations between stations along the same river, and connect the application to live sensor feeds.

8. Conclusion

We built a working water-quality forecasting and decision-support system. On real Indian river data, simple models predicting the yearly change in WQI gave a modest improvement in RMSE over the naive baseline, while the baseline remained competitive in MAE and category accuracy, which is an honest and useful finding about how predictable year-to-year water quality is from yearly averages. On synthetic daily data, the full 7-day forecasting and alert pipeline worked as designed. The two-tab application shows both parts and turns forecasts into plain acceptable, caution and alert messages.

9. References

1. Anbarivan N.L, "Indian water quality data", Kaggle dataset: https://www.kaggle.com/datasets/anbarivan/indian-water-quality-data (accessed 26 October 2026).
2. Anbarivan N.L, "Indian water quality analysis and prediction", Kaggle notebook on the same dataset: https://www.kaggle.com/code/anbarivan/indian-water-quality-analysis-and-prediction (reference for the WQI banding approach).

3. Central Pollution Control Board (CPCB), "Water Quality Criteria" (designated best use classes): https://cpcb.nic.in/water-quality-criteria/ (accessed 26 October 2026).
4. Smart India Hackathon 2024, problem statement SIH1694 (real-time Ganga river water quality forecasting using AI-enabled decision support), as mapped to project P18 in the CSE276 CA1 project list, Lovely Professional University.
5. Software: Python, pandas, NumPy, scikit-learn, XGBoost, statsmodels, matplotlib, Plotly, Streamlit.

Appendix: How to run

1. Create a virtual environment and install the libraries.
2. Synthetic part: run src/generate_data.py, src/preprocess.py, src/wqi.py, src/forecast.py in this order.
3. Real part: place water_dataX.csv in the data folder and run src/real_data_forecast.py.
4. Start the demo with: streamlit run src/app.py

Source code: https://github.com/princesahu500000-cpu/ai-water-quality-forecasting